

Sharp Continuous-Order Comparison for Modified-Bessel Ratios: a Riccati Proof, the Optimal Real-Order Domain, and Lean Verification

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Abstract

For $x > 0$ put $\rho_\alpha(x) = I_{\alpha+1}(x)/I_\alpha(x)$, where I_α is the modified Bessel function of the first kind. We study the sharp continuous-order comparison

$$0 < \rho_\mu(x) - \rho_\nu(x) < \frac{\nu - \mu}{x}.$$

The lower inequality is the classical strict order decrease of the Bessel ratio. For the upper inequality we give a direct two-flow Riccati proof. If $h = \rho_\mu - \rho_\nu$, $b = (\nu - \mu)/x$, and $h = b$ at a contact point, then

$$h' = -\frac{(\nu - \mu)(\mu + \nu + 1)}{x^2}, \quad (h - b)' = -\frac{(\nu - \mu)(\mu + \nu)}{x^2}.$$

The second identity, not the first, is the contact orientation. It proves the bound for $\mu + \nu > 0$. The degenerate boundary $\mu + \nu = 0$ follows from uniqueness for the resulting homogeneous scalar ODE. Conversely, elementary recurrence bounds force a counterexample whenever $\mu + \nu < 0$; neither a K function nor a Bessel asymptotic expansion is needed. Thus, on the natural positivity range $-1 < \mu < \nu$, the upper bound holds for every $x > 0$ if and only if $\mu + \nu \geq 0$.

The nonnegative-order statement itself is known: it is equivalent to Lemma 10 of Freitas–Laugesen. Their proof already contains a first-contact argument, so no novelty is claimed for that mechanism. The contributions here are the explicit sharp ratio formulation, the optimal extension/obstruction across negative orders, an elementary proof without asymptotics, the correction of an earlier unit-step paper statement, and a Lean 4/Mathlib formalization of the complete optimal-domain theorem. The Lean artifact has no `sorry`; its axiom oracle returns exactly `propext`, `Classical.choice`, and `Quot.sound`. We retain the resulting Feynman–Hellmann application: every character sector gap in two-dimensional Wilson $U(1)$ and $SU(2)$ lattice gauge theory decreases strictly with the bare coupling.

1 Introduction

The ratios $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$ of modified Bessel functions of consecutive orders appear throughout applied analysis, probability and statistical physics, and a substantial literature is devoted to bounding them by algebraic functions of ν and x [1, 2, 3, 4, 5, 6]. This note concerns the behaviour under arbitrary real order shifts of the logarithmic derivative

$$\frac{\partial}{\partial x} \log I_\nu(x) = \frac{I'_\nu(x)}{I_\nu(x)}.$$

Theorem 1 (sharp continuous-order comparison). *Let $x > 0$ and $0 \leq \mu < \nu$. Then*

$$0 < \rho_\mu(x) - \rho_\nu(x) < \frac{\nu - \mu}{x}. \quad (1)$$

Equivalently,

$$\frac{I'_\mu(x)}{I_\mu(x)} < \frac{I'_\nu(x)}{I_\nu(x)}. \quad (2)$$

The left inequality is classical; the proof below addresses the sharp right inequality directly.

Theorem 2 (optimal real-order domain). *Let $-1 < \mu < \nu$. The lower inequality in (1) holds for every $x > 0$. The upper inequality holds for every $x > 0$ if and only if*

$$\mu + \nu \geq 0.$$

If $\mu + \nu < 0$, then the upper inequality fails for at least one $x > 0$.

Remark 3 (priority and corrected scope). The nonnegative-order conclusion (2) is Lemma 10 of Freitas and Laugesen [7]. A previous version of this note stated that fractional steps did not follow from its method and described their proof as a Bessel-zero argument. Both statements were wrong. Their published proof defines the logarithmic-derivative difference, uses its positive small- x sign, and excludes a first zero directly from the modified-Bessel differential equation. The two-flow calculation below is the same contact geometry written for ρ_μ and ρ_ν . We therefore claim neither the nonnegative-order theorem nor the first-contact mechanism as new. Segura’s Riccati/nulleline analysis [8] is also close in method, although its stated monotonicity results vary x at fixed order rather than compare two arbitrary orders. We also checked the negative-order individual-ratio bounds of Hornik–Grün [9] and the accessible primary record for the later modified-Lommel monotonicity paper of Mao–Tian [10]; neither source states the cross-order iff boundary below. The candidate contribution isolated here is the optimal negative-order classification in Theorem 2, together with its formal verification. A targeted primary-source search found no prior statement of this exact boundary; that conclusion remains explicitly subject to external priority review.

The purpose of this corrected note is threefold.

First, we expose the entire fractional-order comparison as one Riccati contact identity. This avoids Bessel zeros and high-order uniform asymptotics in the proof region $\mu + \nu > 0$.

Second, we resolve the contact-degenerate line $\mu + \nu = 0$ by ODE uniqueness and prove necessity below the line from the recurrence alone. This produces an exact global domain rather than a sufficient condition, without invoking asymptotics.

Third, we record a consequence in two-dimensional lattice gauge theory (we are not aware of an explicit statement in the literature, though it may well be folklore): by the Feynman–Hellmann identity — exact here, because the 2D transfer operator is diagonal in a β -independent character basis — the continuous-order comparison of Theorem 1 is precisely the strict decrease in the bare coupling β of every mass gap between character sectors, for the $U(1)$ and $SU(2)$ Wilson actions simultaneously (Section 4). We emphasize the modest scope: this is a statement about the exactly soluble two-dimensional theory; nothing here is claimed about four-dimensional gauge theory.

Section 5 presents the machine-checked Lean 4/Mathlib verification of the full optimal-domain theorem (no `sorry`; standard axioms only). Section 6 reports an independent numerical audit.

2 The two-flow Riccati proof and the optimal domain

Write $\delta = \nu - \mu > 0$ and

$$h(x) = \rho_\mu(x) - \rho_\nu(x), \quad b(x) = \frac{\delta}{x}, \quad G(x) = h(x) - b(x).$$

The derivative and recurrence identities imply the Riccati equation

$$\rho'_\alpha = 1 - \rho_\alpha^2 - \frac{2\alpha + 1}{x} \rho_\alpha. \quad (3)$$

Subtracting the equations at μ and ν gives

$$h' = -(\rho_\mu + \rho_\nu)h - \frac{2\mu + 1}{x}h + \frac{2\delta}{x}\rho_\nu. \quad (4)$$

At a point where $h = b = \delta/x$, substitute $\rho_\mu = \rho_\nu + \delta/x$ into (4). Direct cancellation yields

$$h' = -\frac{\delta(\mu + \nu + 1)}{x^2}, \quad G' = h' - b' = -\frac{\delta(\mu + \nu)}{x^2}. \quad (5)$$

This distinction is essential: the first displayed derivative is the derivative of the ratio difference, whereas first-contact orientation is controlled by the second.

For $\alpha > -1$, positivity of the Gamma series and the three-term recurrence give the exact reciprocal identity

$$\frac{1}{\rho_\alpha(x)} = \frac{2(\alpha + 1)}{x} + \rho_{\alpha+1}(x). \quad (6)$$

In particular,

$$0 < \rho_\alpha(x) < \frac{x}{2(\alpha + 1)}, \quad \rho_\alpha(x) > \frac{1}{2(\alpha + 1)/x + x/(2(\alpha + 2))}. \quad (7)$$

These elementary bounds give an explicit interval near zero on which both $h > 0$ and $G < 0$; no asymptotic notation is required.

If $\mu + \nu > 0$ and G had a first zero c , then $G < 0$ immediately to its left but (5) would give $G'(c) < 0$. Differentiability would instead force $G > G(c) = 0$ immediately to the left, a contradiction. Hence $G < 0$ for all $x > 0$. This proves the upper half of Theorem 1 and sufficiency in Theorem 2 away from the boundary. At a putative zero of h , (4) gives

$$h' = \frac{2\delta}{x}\rho_\nu > 0.$$

The same first-contact argument, now using the positive seed from (7), proves $h > 0$ throughout $(-1, \infty)$.

For the boundary and obstruction it is useful to retain the gap rather than specialize at contact. Direct algebra in the two Riccati flows gives the exact linear equation

$$G' = -\left(\rho_\mu + \rho_\nu + \frac{\mu + \nu + 1}{x}\right)G - \frac{\delta(\mu + \nu)}{x^2}. \quad (8)$$

When $\mu + \nu = 0$ this equation is homogeneous. If G vanished after the negative seed, reverse the interval from that zero back to the seed. The coefficient is continuous and bounded there,

so uniqueness for a scalar linear ODE (equivalently, Gronwall's inequality) would force G to vanish identically on the reversed interval, contradicting the strict seed. Thus $G < 0$ also on the boundary.

It remains to prove that the half-plane cannot be enlarged. Suppose $s := \mu + \nu < 0$ and, for contradiction, $G(x) < 0$ for every $x > 0$. From (7), whenever $x^2 \geq 12(\alpha + 1)(\alpha + 2)$,

$$\rho_\alpha(x) > \frac{3(\alpha + 2)}{2x}. \quad (9)$$

Hence, for all sufficiently large x ,

$$A(x) := \rho_\mu(x) + \rho_\nu(x) + \frac{s+1}{x} > \frac{5s+14}{2x} > \frac{2}{x},$$

where the last inequality uses $s > -2$. Put $F = -\delta s > 0$, $H = x^2 G$, and $J = H - Fx$. Equation (8) yields

$$H' = (2x - x^2 A)G + F > F, \quad J' > 0$$

on that tail, because both $2x - x^2 A$ and G are negative. Choose a tail point X and then $z = X - H(X)/F + 1$. Monotonicity of J gives

$$H(z) \geq H(X) + F(z - X) = F > 0,$$

contradicting $H(z) = z^2 G(z) < 0$. Therefore the upper bound fails at some positive argument whenever $\mu + \nu < 0$, completing Theorem 2 without asymptotics.

For a concrete audit witness, at 100 decimal digits,

$$(\mu, \nu, x) = (-0.8, -0.4, 10), \quad \rho_\mu - \rho_\nu = 0.042682961675102858\dots > 0.04 = \frac{\nu - \mu}{x}.$$

3 The unit-step Amos calibration

Throughout, $x > 0$, $\nu \geq 0$, I_ν is the modified Bessel function of the first kind, all values $I_\nu(x)$ are strictly positive, and $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x) \in (0, 1)$. We use two classical facts, valid for all real orders (see [15, 10.29.1–10.29.2]):

$$I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu}{x} I_\nu(x), \quad (10)$$

$$I'_\nu(x) = \frac{1}{2}(I_{\nu-1}(x) + I_{\nu+1}(x)). \quad (11)$$

Specializing Theorem 1 to the unit shift gives

$$\frac{I'_{\nu+1}(x)}{I_{\nu+1}(x)} - \frac{I'_\nu(x)}{I_\nu(x)} = \frac{1}{x} - (\rho_\nu(x) - \rho_{\nu+1}(x)) > 0. \quad (12)$$

The older four-step proof of this specialization is retained because it records the exact equivalence with the Amos bound.

Step 1: a closed form for the logarithmic derivative

Dividing (11) by I_ν and using (10) in the form $I_{\nu-1}/I_\nu = \rho_\nu + 2\nu/x$:

$$\frac{I'_\nu(x)}{I_\nu(x)} = \frac{1}{2} \left(\frac{I_{\nu-1}}{I_\nu} + \frac{I_{\nu+1}}{I_\nu} \right) = \frac{1}{2} \left(\rho_\nu + \frac{2\nu}{x} + \rho_\nu \right) = \rho_\nu(x) + \frac{\nu}{x}. \quad (13)$$

Step 2: reduction to a ratio inequality

By (13), for the unit step $\nu \rightarrow \nu + 1$,

$$\frac{I'_{\nu+1}}{I_{\nu+1}} - \frac{I'_\nu}{I_\nu} = \rho_{\nu+1} - \rho_\nu + \frac{1}{x},$$

which is (12). Writing the recurrence (10) at order $\nu + 1$ as $\rho_{\nu+1} = 1/\rho_\nu - 2(\nu + 1)/x$, the inequality $\rho_\nu - \rho_{\nu+1} < 1/x$ is equivalent to

$$\frac{1}{\rho_\nu(x)} - \rho_\nu(x) > \frac{2\nu + 1}{x}. \quad (14)$$

Step 3: the Amos-type bound

We use the classical upper bound (Amos [1]; proved best of its form and valid for all $\nu \geq 0$ by Segura [5, Thm. 2], after the index shift $\nu \mapsto \nu + 1$):

$$\rho_\nu(x) < U_\nu(x) := \frac{x}{a + s}, \quad a = \nu + \frac{1}{2}, \quad s = \sqrt{a^2 + x^2}. \quad (15)$$

Step 4: the exact calibration

Lemma 4 (calibration). *Let $a, x > 0$ and $s = \sqrt{a^2 + x^2}$, $U = x/(a + s)$. Then*

$$\frac{1}{U} - U = \frac{2a}{x},$$

and for every t with $0 < t < U$,

$$\frac{1}{t} - t > \frac{2a}{x}.$$

Proof. Since $(s - a)(s + a) = s^2 - a^2 = x^2$, we have $U = x/(a + s) = (s - a)/x$, hence

$$\frac{1}{U} - U = \frac{a + s}{x} - \frac{s - a}{x} = \frac{2a}{x}.$$

The map $t \mapsto 1/t - t$ is strictly decreasing on $(0, \infty)$, so $t < U$ gives $1/t - t > 1/U - U = 2a/x$. $\square$

Applying Lemma 4 with $t = \rho_\nu(x) < U_\nu(x)$ and $a = \nu + \frac{1}{2}$ yields (14) strictly, hence (12) and the unit-shift instance of (1). $\square$

Remark 5 (equivalence with the Amos bound). The chain of Step 2 consists of equivalences: since $t \mapsto 1/t - t$ is a strictly decreasing bijection of $(0, \infty)$ onto $\mathbb{R}$, and U_ν is precisely the positive root of $1/t - t = (2\nu + 1)/x$ (Lemma 4), the inequality (14) is *equivalent* to $\rho_\nu < U_\nu$. Therefore: *the unit-step order-monotonicity (12) is equivalent to the classical Amos upper bound (15)*. Any of the many proofs of the latter [1, 3, 4, 5] yields the former, and conversely the continuous monotonicity of Freitas–Laugesen [7] yields, in particular, the Amos bound. Their proof of order monotonicity is the first-contact argument described in Remark 3, not a Bessel-zero argument. The bound U_ν is also recognizable as the b_0 -type bound of [3]: the positive root of the Riccati-derived quadratic $t^2 + \frac{2\nu+1}{x}t - 1 = 0$.

4 Application: bare-coupling monotonicity of all 2D sector gaps

Consider lattice gauge theory on a two-dimensional rectangular lattice with Wilson action and compact structure group G , at bare coupling $\beta > 0$. We fix notation carefully, separating the three objects involved.

(a) *Character coefficients.* The one-plaquette Boltzmann weight expands as (see, e.g., [16, 17])

$$e^{\beta \operatorname{Re} \chi_f(U)} = \sum_r d_r c_r(\beta) \chi_r(U), \quad c_r(\beta) = \frac{1}{d_r} \int_G \overline{\chi_r(U)} e^{\beta \operatorname{Re} \chi_f(U)} dU,$$

the sum over irreducible representations r of G , $d_r = \chi_r(e)$. For the two classical cases ($n \in \mathbb{Z}_{\geq 0}$, $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$; verified independently to 30 digits in the audit of Section 6):

$$G = U(1) : \quad c_n(\beta) = I_n(\beta), \quad G = SU(2) : \quad c_j(\beta) = \frac{I_{2j+1}(2\beta)}{\beta},$$

the $SU(2)$ formula being the statement that the Haar class integral equals $\int_G \chi_j(U) e^{\beta \chi_f(U)} dU = (2j+1) I_{2j+1}(2\beta)/\beta$, i.e. $d_j c_j(\beta)$ with $d_j = 2j+1$ (all $SU(2)$ characters are real, so $\operatorname{Re} \chi_f = \chi_f = 2 \cos \theta$).

(b) *Transfer eigenvalues.* In two dimensions, gauge fixing reduces the theory to a product of independent plaquette weights, and the transfer operator (one lattice row, temporal gauge) is diagonal in the character basis $\{\chi_r\}$ — the standard exact solubility of 2D lattice gauge theory [16, 13]; see [14] for a textbook treatment of the character expansion — a basis independent of β — with eigenvalue in sector r proportional to $c_r(\beta)$, the proportionality constant being a sector-independent normalization. Ratios of transfer eigenvalues between sectors are therefore exactly $c_r(\beta)/c_{r'}(\beta)$, with every normalization convention cancelling.

(c) *Sector gaps.* For two sectors with $c_r(\beta) > c_{r'}(\beta) > 0$ (in both models the coefficient is strictly decreasing in the sector label, so “heavier” sectors have smaller coefficients), the mass gap of sector r' relative to r is

$$m_{r,r'}(\beta) = \log \frac{c_r(\beta)}{c_{r'}(\beta)} > 0,$$

e.g. for $U(1)$ the fundamental string tension is $m_{0,1} = \log(I_0/I_1)$ and the $v87$ bracket instance is $m_{1,2} = \log(I_1/I_2)$. Because the eigenbasis is β -independent, $\frac{d}{d\beta} m_{r,r'}$ acts on the eigenvalues alone — the Feynman–Hellmann identity in its exact, diagonal form; no perturbation theory is involved.

Corollary 6. *For 2D Wilson lattice gauge theory:*

(i) $G = U(1)$: for all integers $0 \leq n < n'$,

$$\frac{d}{d\beta} m_{n,n'}(\beta) = \frac{d}{d\beta} \log \frac{I_n(\beta)}{I_{n'}(\beta)} = (\rho_n - \rho_{n'})'(\beta) - \frac{n' - n}{\beta} < 0.$$

Every sector gap — in particular the string tension $\log(I_0/I_1)$ and the bracket instance $\log(I_1/I_2)$ — is strictly decreasing in β .

(ii) $G = SU(2)$: for all $0 \leq j < j'$ in $\frac{1}{2}\mathbb{Z}$,

$$\frac{d}{d\beta} m_{j,j'}(\beta) = 2 \left[(\log I_{2j+1})'(2\beta) - (\log I_{2j'+1})'(2\beta) \right] < 0,$$

the $1/\beta$ prefactors cancelling in the ratio $c_j/c_{j'} = I_{2j+1}(2\beta)/I_{2j'+1}(2\beta)$, which is free of representation-dimension factors.

Both assertions are direct instances of Theorem 1; the orders happen to be integer-spaced, so the older unit-step proof in Section 3 would also suffice.

Proof. (i) By (13), $\frac{d}{d\beta} \log(I_n/I_{n'}) = (\rho_n + n/\beta) - (\rho_{n'} + n'/\beta) = (\rho_n - \rho_{n'}) - (n' - n)/\beta$, and (1) gives $\rho_n - \rho_{n'} < (n' - n)/\beta$ strictly. (ii) Identical, at argument $x = 2\beta$ and orders $2j+1 < 2j'+1$; the chain rule contributes the factor 2. $\square$

Remark 7 (scope). Corollary 6 is a statement about the exactly soluble two-dimensional theory. The analogous monotonicity $m'(\beta) \leq 0$ for the four-dimensional mass gap on a weak-to-strong coupling corridor is an open problem (universally expected in lattice physics, covered by no theorem); the present corollary is the exact 2D instance of that bracket and nothing more. In the terminology of the audit-first programme to which this note belongs, the 2D instance is *signed*; the 4D bracket remains a named hypothesis.

5 The machine-checked algebraic core

The full optimal-domain theorem is formalized in the C4 modules `BesselNegative.lean` and `FractionalOrderOptimal.lean`. The first extends the actual Gamma-series definition, positivity, recurrence, derivative identity, and Riccati equation from nonnegative order to the natural strip $\nu > -1$. The second checks the strict order decrease, the positive-sum first-contact barrier, the homogeneous-ODE boundary argument, and the recurrence-only negative-sum obstruction. Both files contain no `sorry` and introduce no project axiom. The principal checked endpoints are:

```
theorem besselRatioReal_fractional_upper_all_iff
  (mu nu : Real) (hmu : -1 < mu) (hmu_nu : mu < nu) :
  (forall x, 0 < x ->
    besselRatioReal mu x - besselRatioReal nu x
    < (nu - mu) / x) <-> 0 <= mu + nu

theorem besselRatioReal_fractional_two_sided
  (mu nu : Real) (hmu : -1 < mu) (hmu_nu : mu < nu)
  (hsum : 0 <= mu + nu) {x : Real} (hx : 0 < x) :
  0 < besselRatioReal mu x - besselRatioReal nu x /\
  besselRatioReal mu x - besselRatioReal nu x
  < (nu - mu) / x
```

The obstruction endpoint is existential: if $\mu + \nu < 0$, Lean produces an $x > 0$ at which the upper inequality fails. The implementation also exposes the full signed-gap ODE and the exact contact derivative. The axiom oracle for the registered headline declarations returns exactly:

`[propext, Classical.choice, Quot.sound]`

Verification used Lean 4.29.0-rc6 and pinned Mathlib commit

07642720480157414db592fa85b626dafb71355b.

On 4 August 2026 the modules were built in a Colab Pro+ CPU/high-memory runtime: 8167 Lake jobs completed for the optimal module, its `.olean` was materialized, and the oracle file was compiled. Runtime opening and teardown were recorded; the runtime was disconnected and deleted after 31 minutes.

Honest scope. The machine-checked theorem now covers both strict inequalities on the optimal strip, including the negative-order boundary and obstruction in Theorem 2. The priority claim is not machine-checkable and remains a candidate pending an independent bibliographic review. Source presence is not reported as verification: the claim above refers to the materialized Colab builds and their oracle output.

6 Numerical audit

The fractional-order statement was attacked independently at 100 decimal digits on 800 deterministic adversarial points (seed 490060), including tiny order separations, values near $\mu + \nu = 0$, and arguments from the small- to large- x regimes. No failure occurred in the valid domain $\mu + \nu \geq 0$; this is diagnostic, not proof. The same script verifies the explicit invalid-domain witness in Section 2. In addition, the older unit-step/application inequalities were checked at 50 decimal digits (mpmath), on the grid $\beta \in \{10^{-2}, \dots, 10^4\}$ (16 values, geometrically spread), $\nu \in \{0, 1, \dots, 29\} \cup \{50, 100, 500\}$: the closed form (13) holds to $1.4 \cdot 10^{-18}$ (central finite difference at step 10^{-20}); the Amos margin $U_\nu - \rho_\nu$ is strictly positive throughout, with minimum $2.5 \cdot 10^{-9}$ attained at $(\beta, \nu) = (10^4, 0)$ (the bound is asymptotically sharp, as it must be, being extremal of its form); the quantitative slack $1/\beta - (\rho_\nu - \rho_{\nu+1})$ is strictly positive throughout, with minimum $1.5 \cdot 10^{-8}$ at $(10^4, 1)$; and $m(\beta) = \log(I_1/I_2)$ has $m'(\beta) < 0$ at every grid point, ranging from -1.96 at $\beta = \frac{1}{2}$ to $-1.9 \cdot 10^{-5}$ at $\beta = 278$. The $SU(2)$ Haar class integral $\int_G \chi_j e^{\beta \chi_f} dU = \frac{2}{\pi} \int_0^\pi \sin \theta \sin((2j+1)\theta) e^{2\beta \cos \theta} d\theta$ was verified to equal $(2j+1) I_{2j+1}(2\beta)/\beta$ to 30 digits at $j \in \{0, 1, 2\}$, $\beta \in \{0.7, 2\}$ — hence $c_j = I_{2j+1}(2\beta)/\beta$ after dividing by $d_j = 2j+1$. The verification script and the sources of this note are archived in the programme repository:

[https:](https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh)

[//github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh](https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh)

Version and audit note

This note is part of an audit-first research programme whose records, Lean sources and numerical suites are public at <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME> (machine-checked core) and the satellite repositories linked there. The author thanks the adversarial-audit process itself. It first located [7], correcting a novelty claim, and later exposed a second error in version 6: the paper said its method did not close fractional steps and misdescribed Lemma 10 as a Bessel-zero proof. The contact calculation (5) corrects that statement. This version separates the known nonnegative-order theorem and its known first-contact route from the candidate contribution: the optimal negative-order classification, now formalized end to end. Numerical residuals and a finite mesh are never used as proof.

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